



National Textile University

Department of Computer Science

Subject:

Operating System

Submitted to:

Nasir Mahmood

Submitted by:

Afia Maham

Reg number:

23-NTU-CS-1126

Lab no:

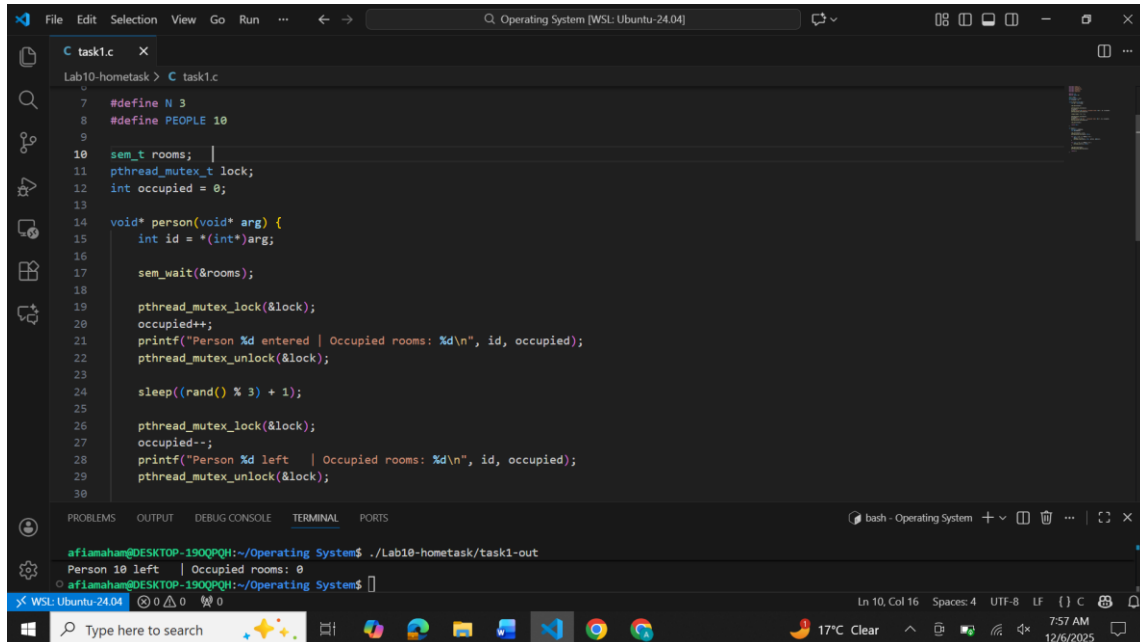
10 - hometask

Semester:

5th

Task 1:

Code:

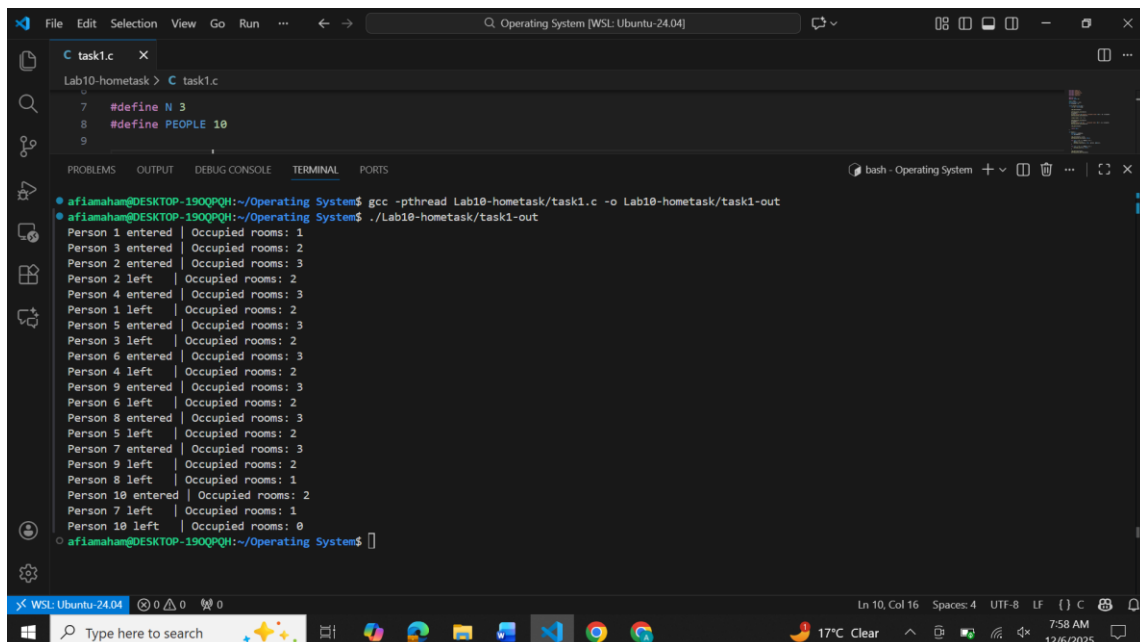


```
File Edit Selection View Go Run ... Operating System [WSL: Ubuntu-24.04]
C task1.c
Lab10-hometask > C task1.c
7 #define N 3
8 #define PEOPLE 10
9
10 sem_t rooms;
11 pthread_mutex_t lock;
12 int occupied = 0;
13
14 void* person(void* arg) {
15     int id = *(int*)arg;
16
17     sem_wait(&rooms);
18
19     pthread_mutex_lock(&lock);
20     occupied++;
21     printf("Person %d entered | Occupied rooms: %d\n", id, occupied);
22     pthread_mutex_unlock(&lock);
23
24     sleep((rand() % 3) + 1);
25
26     pthread_mutex_lock(&lock);
27     occupied--;
28     printf("Person %d left | Occupied rooms: %d\n", id, occupied);
29     pthread_mutex_unlock(&lock);
30
31 }
32
33 int main() {
34     int ids[PEOPLE];
35     for (int i = 0; i < PEOPLE; i++) {
36         ids[i] = i + 1;
37     }
38     pthread_t threads[PEOPLE];
39     for (int i = 0; i < PEOPLE; i++) {
40         pthread_create(&threads[i], NULL, person, &ids[i]);
41     }
42     for (int i = 0; i < PEOPLE; i++) {
43         pthread_join(threads[i], NULL);
44     }
45     return 0;
46 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
bash - Operating System
afianaham@DESKTOP-190QPQH:~/Operating System$ ./Lab10-hometask/task1-out
Person 10 left | Occupied rooms: 0
afianaham@DESKTOP-190QPQH:~/Operating System$
```

Output:



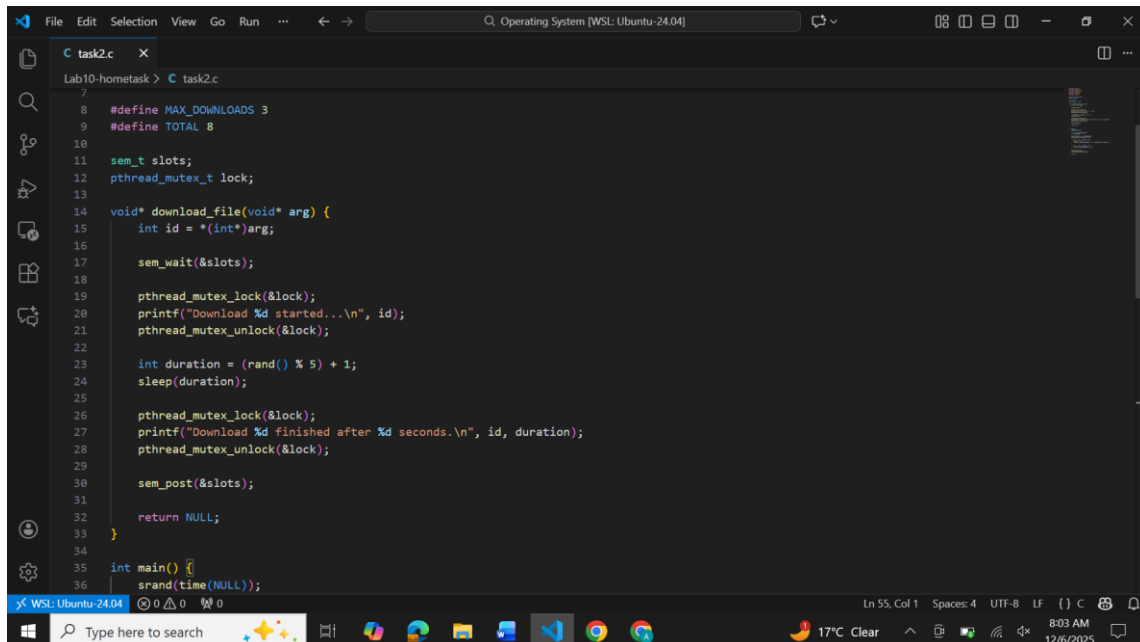
```
File Edit Selection View Go Run ... Operating System [WSL: Ubuntu-24.04]
C task1.c
Lab10-hometask > C task1.c
7 #define N 3
8 #define PEOPLE 10
9
10 sem_t rooms;
11 pthread_mutex_t lock;
12 int occupied = 0;
13
14 void* person(void* arg) {
15     int id = *(int*)arg;
16
17     sem_wait(&rooms);
18
19     pthread_mutex_lock(&lock);
20     occupied++;
21     printf("Person %d entered | Occupied rooms: %d\n", id, occupied);
22     pthread_mutex_unlock(&lock);
23
24     sleep((rand() % 3) + 1);
25
26     pthread_mutex_lock(&lock);
27     occupied--;
28     printf("Person %d left | Occupied rooms: %d\n", id, occupied);
29     pthread_mutex_unlock(&lock);
30
31 }
32
33 int main() {
34     int ids[PEOPLE];
35     for (int i = 0; i < PEOPLE; i++) {
36         ids[i] = i + 1;
37     }
38     pthread_t threads[PEOPLE];
39     for (int i = 0; i < PEOPLE; i++) {
40         pthread_create(&threads[i], NULL, person, &ids[i]);
41     }
42     for (int i = 0; i < PEOPLE; i++) {
43         pthread_join(threads[i], NULL);
44     }
45     return 0;
46 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
bash - Operating System
afianaham@DESKTOP-190QPQH:~/Operating System$ gcc -pthread Lab10-hometask/task1.c -o Lab10-hometask/task1-out
afianaham@DESKTOP-190QPQH:~/Operating System$ ./Lab10-hometask/task1-out
Person 1 entered | Occupied rooms: 1
Person 3 entered | Occupied rooms: 2
Person 2 entered | Occupied rooms: 3
Person 2 left | Occupied rooms: 2
Person 4 entered | Occupied rooms: 3
Person 1 left | Occupied rooms: 2
Person 5 entered | Occupied rooms: 3
Person 3 left | Occupied rooms: 2
Person 6 entered | Occupied rooms: 3
Person 4 left | Occupied rooms: 2
Person 9 entered | Occupied rooms: 3
Person 6 left | Occupied rooms: 2
Person 8 entered | Occupied rooms: 3
Person 5 left | Occupied rooms: 2
Person 7 entered | Occupied rooms: 3
Person 9 left | Occupied rooms: 2
Person 8 left | Occupied rooms: 1
Person 10 entered | Occupied rooms: 2
Person 7 left | Occupied rooms: 1
Person 10 left | Occupied rooms: 0
afianaham@DESKTOP-190QPQH:~/Operating System$
```

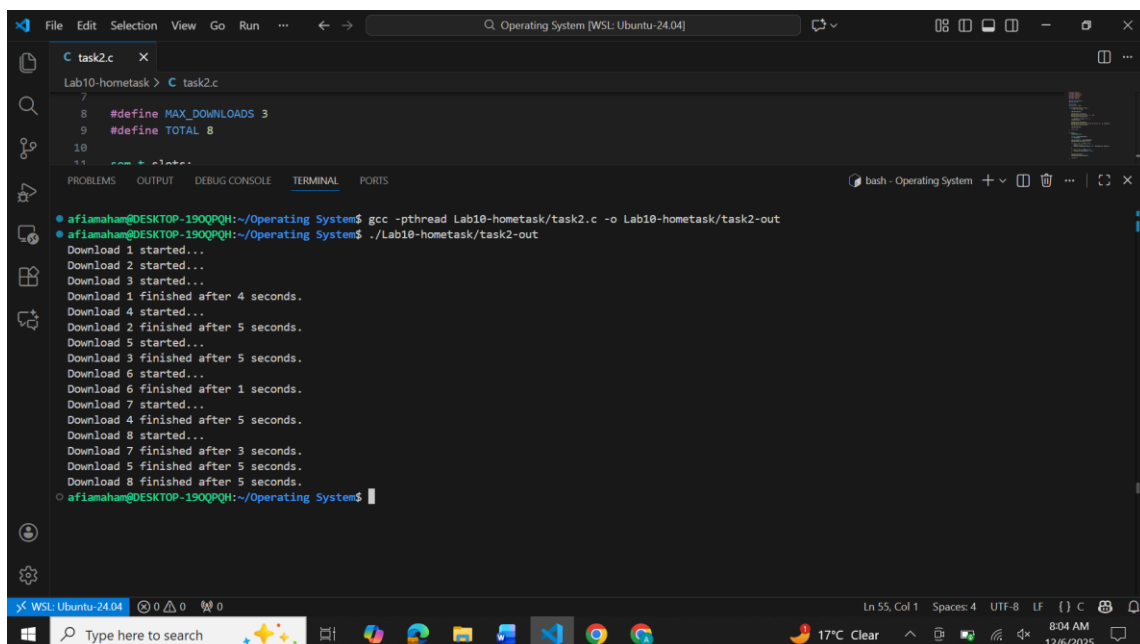
Task 2:

Code:



```
7
8 #define MAX_DOWNLOADS 3
9 #define TOTAL 8
10
11 sem_t slots;
12 pthread_mutex_t lock;
13
14 void* download_file(void* arg) {
15     int id = *(int*)arg;
16
17     sem_wait(&slots);
18
19     pthread_mutex_lock(&lock);
20     printf("Download %d started...\n", id);
21     pthread_mutex_unlock(&lock);
22
23     int duration = (rand() % 5) + 1;
24     sleep(duration);
25
26     pthread_mutex_lock(&lock);
27     printf("Download %d finished after %d seconds.\n", id, duration);
28     pthread_mutex_unlock(&lock);
29
30     sem_post(&slots);
31
32     return NULL;
33 }
34
35 int main() {
36     srand(time(NULL));
```

Output:

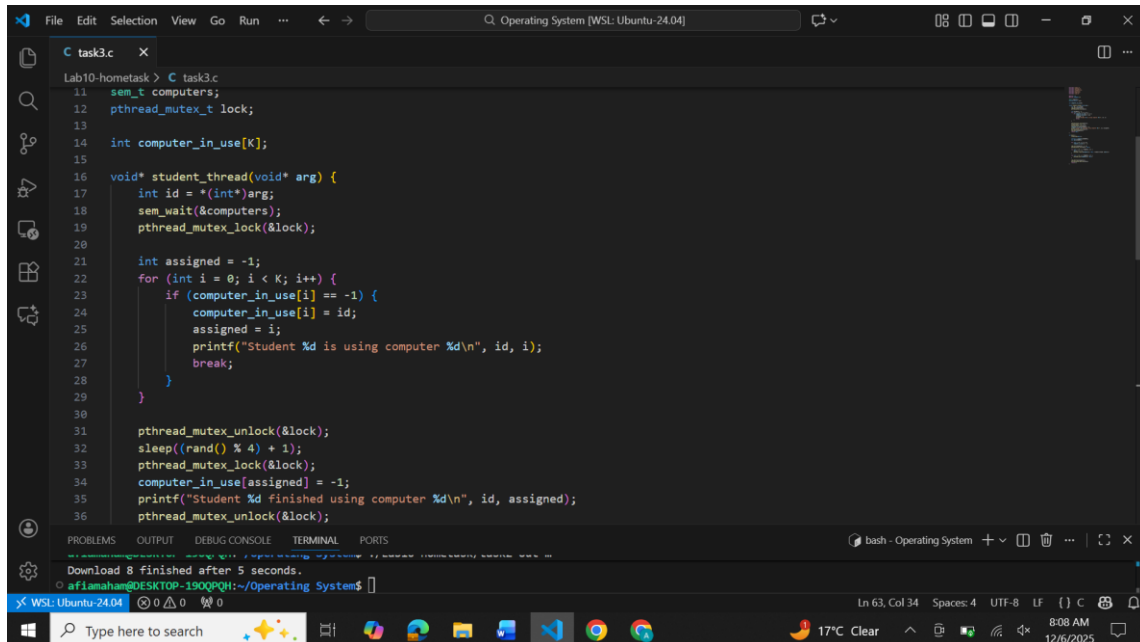


```
Lab10-hometask > C task2.c
7
8 #define MAX_DOWNLOADS 3
9 #define TOTAL 8
10
11 sem_t slots;
12 pthread_mutex_t lock;
13
14 void* download_file(void* arg) {
15     int id = *(int*)arg;
16
17     sem_wait(&slots);
18
19     pthread_mutex_lock(&lock);
20     printf("Download %d started...\n", id);
21     pthread_mutex_unlock(&lock);
22
23     int duration = (rand() % 5) + 1;
24     sleep(duration);
25
26     pthread_mutex_lock(&lock);
27     printf("Download %d finished after %d seconds.\n", id, duration);
28     pthread_mutex_unlock(&lock);
29
30     sem_post(&slots);
31
32     return NULL;
33 }
34
35 int main() {
36     srand(time(NULL));
37     int ids[TOTAL];
38     pthread_t threads[MAX_DOWNLOADS];
39     int i;
40
41     for (i = 0; i < TOTAL; i++) {
42         ids[i] = i;
43     }
44
45     for (i = 0; i < MAX_DOWNLOADS; i++) {
46         pthread_create(&threads[i], NULL, download_file, &ids[i]);
47     }
48
49     for (i = 0; i < MAX_DOWNLOADS; i++) {
50         pthread_join(threads[i], NULL);
51     }
52
53     return 0;
54 }
```

```
afianaham@DESKTOP-190QPQH:~/Operating System$ gcc -pthread Lab10-hometask/task2.c -o Lab10-hometask/task2-out
afianaham@DESKTOP-190QPQH:~/Operating System$ ./Lab10-hometask/task2-out
Download 1 started...
Download 2 started...
Download 3 started...
Download 1 finished after 4 seconds.
Download 4 started...
Download 2 finished after 5 seconds.
Download 5 started...
Download 3 finished after 5 seconds.
Download 6 started...
Download 6 finished after 1 seconds.
Download 7 started...
Download 4 finished after 5 seconds.
Download 8 started...
Download 7 finished after 3 seconds.
Download 5 finished after 5 seconds.
Download 8 finished after 5 seconds.
afianaham@DESKTOP-190QPQH:~/Operating System$
```

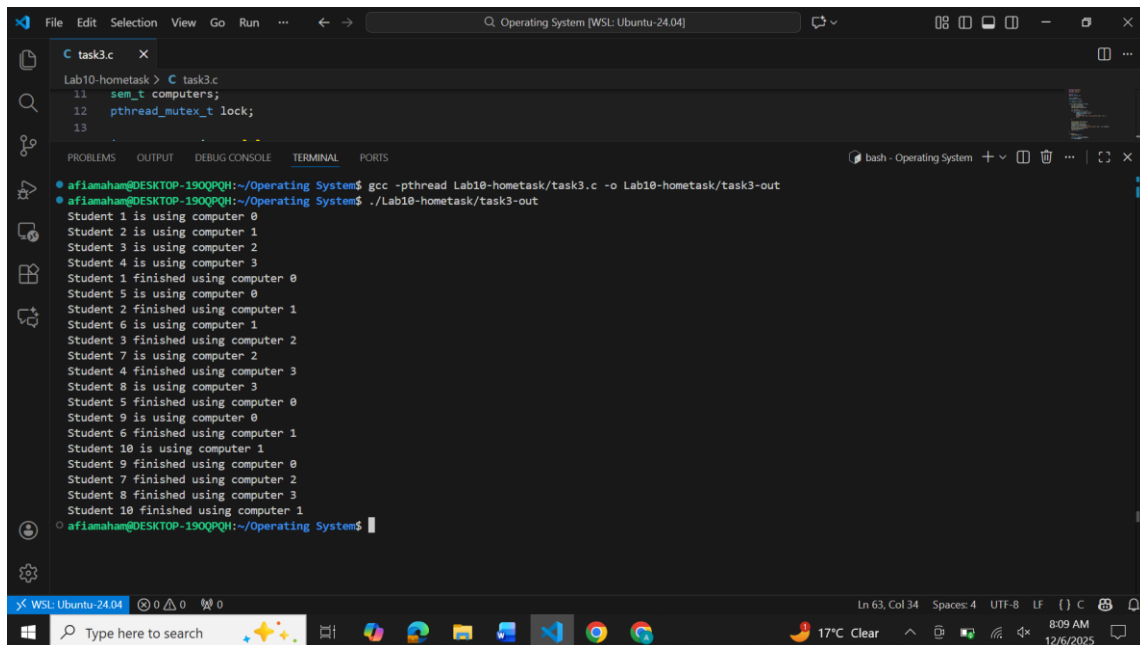
Task 3:

Code:



```
11 sem_t computers;
12 pthread_mutex_t lock;
13
14 int computer_in_use[K];
15
16 void* student_thread(void* arg) {
17     int id = *(int*)arg;
18     sem_wait(&computers);
19     pthread_mutex_lock(&lock);
20
21     int assigned = -1;
22     for (int i = 0; i < K; i++) {
23         if (computer_in_use[i] == -1) {
24             computer_in_use[i] = id;
25             assigned = i;
26             printf("Student %d is using computer %d\n", id, i);
27             break;
28         }
29     }
30
31     pthread_mutex_unlock(&lock);
32     sleep((rand() % 4) + 1);
33     pthread_mutex_lock(&lock);
34     computer_in_use[assigned] = -1;
35     printf("Student %d finished using computer %d\n", id, assigned);
36     pthread_mutex_unlock(&lock);
37 }
```

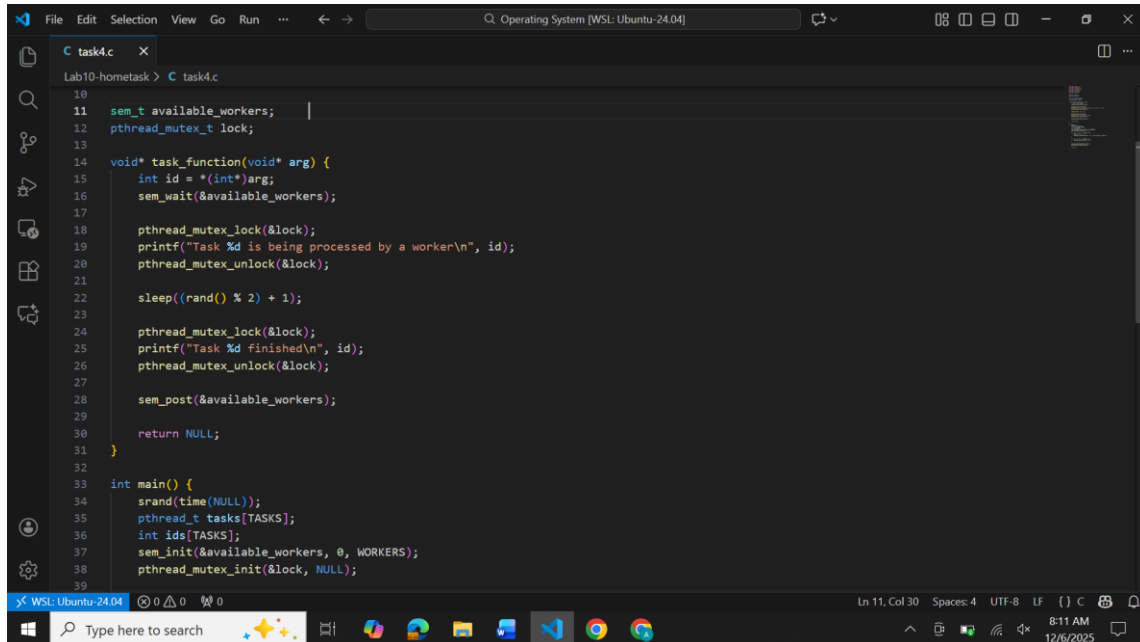
Output:



```
afiamah@DESKTOP-190QPQH:~/Operating System$ gcc -pthread Lab10-hometask/task3.c -o Lab10-hometask/task3-out
afiamah@DESKTOP-190QPQH:~/Operating System$ ./Lab10-hometask/task3-out
Student 1 is using computer 0
Student 2 is using computer 1
Student 3 is using computer 2
Student 4 is using computer 3
Student 1 finished using computer 0
Student 5 is using computer 0
Student 2 finished using computer 1
Student 6 is using computer 1
Student 3 finished using computer 2
Student 7 is using computer 2
Student 4 finished using computer 3
Student 8 is using computer 3
Student 5 finished using computer 0
Student 9 is using computer 0
Student 6 finished using computer 1
Student 10 is using computer 1
Student 9 finished using computer 0
Student 7 finished using computer 2
Student 8 finished using computer 3
Student 10 finished using computer 1
afiamah@DESKTOP-190QPQH:~/Operating System$
```

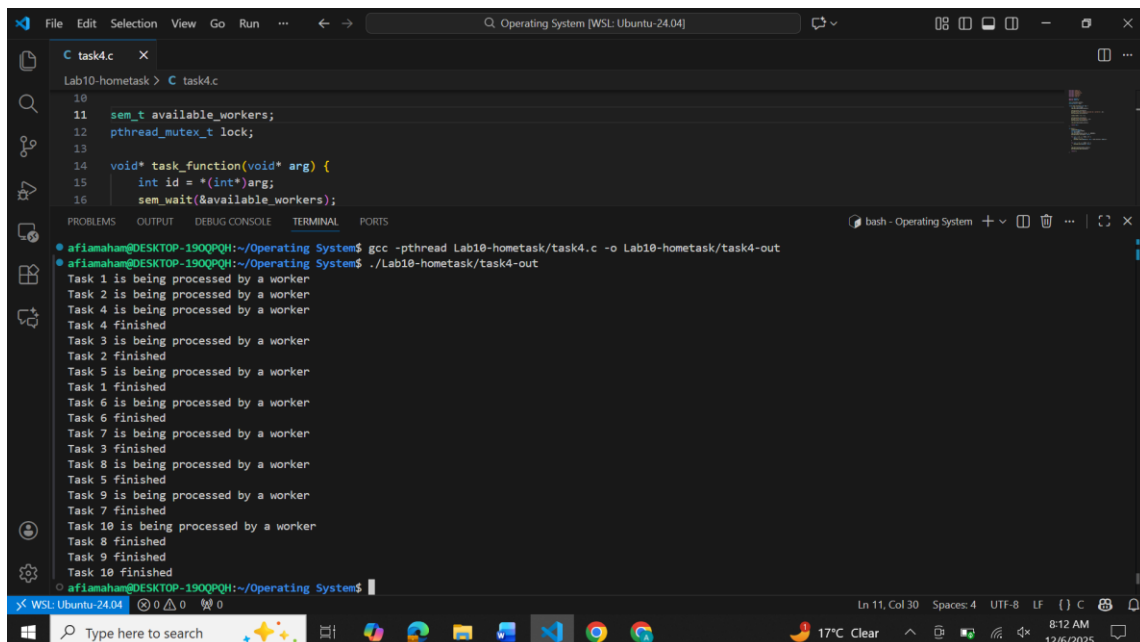
Task 4:

Code:



```
10
11 sem_t available_workers;
12 pthread_mutex_t lock;
13
14 void* task_function(void* arg) {
15     int id = *(int*)arg;
16     sem_wait(&available_workers);
17
18     pthread_mutex_lock(&lock);
19     printf("Task %d is being processed by a worker\n", id);
20     pthread_mutex_unlock(&lock);
21
22     sleep((rand() % 2) + 1);
23
24     pthread_mutex_lock(&lock);
25     printf("Task %d finished\n", id);
26     pthread_mutex_unlock(&lock);
27
28     sem_post(&available_workers);
29
30     return NULL;
31 }
32
33 int main() {
34     srand(time(NULL));
35     pthread_t tasks[TASKS];
36     int ids[TASKS];
37     sem_init(&available_workers, 0, WORKERS);
38     pthread_mutex_init(&lock, NULL);
39 }
```

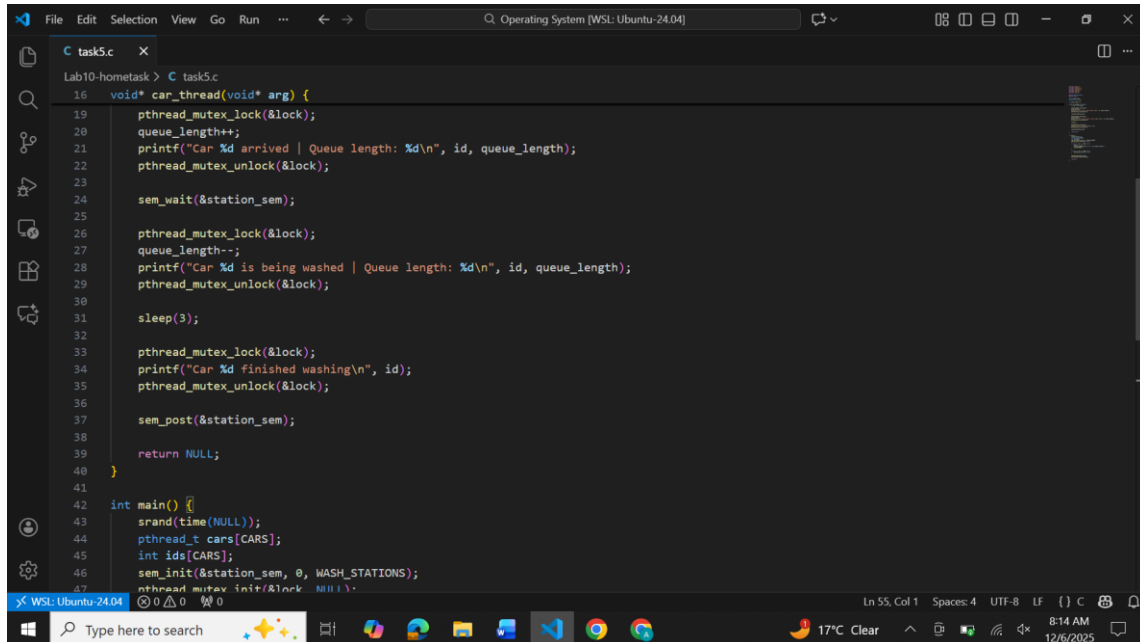
Output:



```
afiamaham@DESKTOP-190QPQH:~/Operating System$ gcc -pthread Lab10-hometask/task4.c -o Lab10-hometask/task4-out
afiamaham@DESKTOP-190QPQH:~/Operating System$ ./Lab10-hometask/task4-out
Task 1 is being processed by a worker
Task 2 is being processed by a worker
Task 4 is being processed by a worker
Task 4 finished
Task 3 is being processed by a worker
Task 2 finished
Task 5 is being processed by a worker
Task 1 finished
Task 6 is being processed by a worker
Task 6 finished
Task 7 is being processed by a worker
Task 3 finished
Task 8 is being processed by a worker
Task 5 finished
Task 9 is being processed by a worker
Task 7 finished
Task 10 is being processed by a worker
Task 8 finished
Task 9 finished
Task 10 finished
afiamaham@DESKTOP-190QPQH:~/Operating System$
```

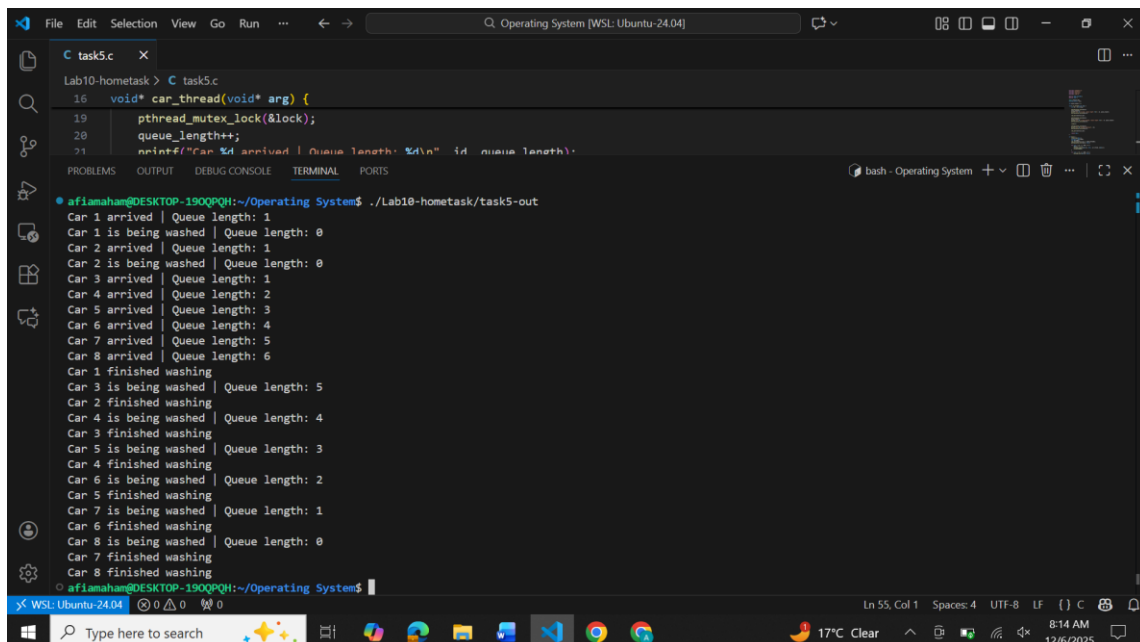
Task 5:

Code:



```
16 void* car_thread(void* arg) {
19     pthread_mutex_lock(&lock);
20     queue_length++;
21     printf("Car %d arrived | Queue length: %d\n", id, queue_length);
22     pthread_mutex_unlock(&lock);
23
24     sem_wait(&station_sem);
25
26     pthread_mutex_lock(&lock);
27     queue_length--;
28     printf("Car %d is being washed | Queue length: %d\n", id, queue_length);
29     pthread_mutex_unlock(&lock);
30
31     sleep(3);
32
33     pthread_mutex_lock(&lock);
34     printf("Car %d finished washing\n", id);
35     pthread_mutex_unlock(&lock);
36
37     sem_post(&station_sem);
38
39     return NULL;
40 }
41
42 int main() {
43     srand(time(NULL));
44     pthread_t cars[CARS];
45     int ids[CARS];
46     sem_init(&station_sem, 0, WASH_STATIONS);
47     pthread_mutex_init(&lock, NULL);
```

Output:



```
afianaham@DESKTOP-190QPQH:~/Operating System$ ./Lab10-hometask/task5-out
Car 1 arrived | Queue length: 1
Car 1 is being washed | Queue length: 0
Car 2 arrived | Queue length: 1
Car 2 is being washed | Queue length: 0
Car 3 arrived | Queue length: 1
Car 4 arrived | Queue length: 2
Car 5 arrived | Queue length: 3
Car 6 arrived | Queue length: 4
Car 7 arrived | Queue length: 5
Car 8 arrived | Queue length: 6
Car 1 finished washing
Car 3 is being washed | Queue length: 5
Car 2 finished washing
Car 4 is being washed | Queue length: 4
Car 3 finished washing
Car 5 is being washed | Queue length: 3
Car 4 finished washing
Car 6 is being washed | Queue length: 2
Car 5 finished washing
Car 7 is being washed | Queue length: 1
Car 6 finished washing
Car 8 is being washed | Queue length: 0
Car 7 finished washing
Car 8 finished washing
```

Task 6:

Code:

```
File Edit Selection View Go Run ... Operating System [WSL: Ubuntu-24.04]
C task6.c
Lab10-hometask > C task6.c
14 void* car_thread(void* arg) {
15     printf("Car %d arrived at the bridge\n", id);
16     pthread_mutex_unlock(&print_lock);
17
18     sem_wait(&bridge_sem);
19
20     pthread_mutex_lock(&print_lock);
21     printf("Car %d is crossing the bridge\n", id);
22     pthread_mutex_unlock(&print_lock);
23
24     int cross_time = (rand() % 4) + 1;
25     sleep(cross_time);
26
27     pthread_mutex_lock(&print_lock);
28     printf("Car %d finished crossing in %d sec\n", id, cross_time);
29     pthread_mutex_unlock(&print_lock);
30
31     sem_post(&bridge_sem);
32
33     return NULL;
34 }
35
36 int main() {
37     srand(time(NULL));
38     pthread_t cars[TOTAL_CARS];
39     int ids[TOTAL_CARS];
40     sem_init(&bridge_sem, 0, MAX_ON_BRIDGE);
41     pthread_mutex_init(&print_lock, NULL);
42     for (int i = 0; i < TOTAL_CARS; i++) {
43
44     }
45 }
46
```

Output:

```
File Edit Selection View Go Run ... Operating System [WSL: Ubuntu-24.04]
C task6.c
Lab10-hometask > C task6.c
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - Operating System
afianaham@DESKTOP-190QPQH:~/Operating System$ ./Lab10-hometask/task6-out
Car 7 arrived at the bridge
Car 1 finished crossing in 1 sec
Car 4 is crossing the bridge
Car 8 arrived at the bridge
Car 2 finished crossing in 1 sec
Car 5 is crossing the bridge
Car 9 arrived at the bridge
Car 10 arrived at the bridge
Car 11 arrived at the bridge
Car 12 arrived at the bridge
Car 4 finished crossing in 1 sec
Car 6 is crossing the bridge
Car 5 finished crossing in 2 sec
Car 7 is crossing the bridge
Car 3 finished crossing in 3 sec
Car 8 is crossing the bridge
Car 6 finished crossing in 3 sec
Car 9 is crossing the bridge
Car 8 finished crossing in 2 sec
Car 10 is crossing the bridge
Car 7 finished crossing in 3 sec
Car 11 is crossing the bridge
Car 10 finished crossing in 1 sec
Car 12 is crossing the bridge
Car 9 finished crossing in 2 sec
Car 12 finished crossing in 2 sec
Car 11 finished crossing in 4 sec
afianaham@DESKTOP-190QPQH:~/Operating System$
```